

Verra VM0042 — Biochar Methodology

Quantifying GHG Emission Reductions from Biochar Production and Application

Reading 4 · Module 3 · Meer Foundation EduSuTo

Course: Climate Change: Carbon Capture and Storage

About Verra and the VM0042 Methodology

Verra is the world's leading voluntary carbon standard organisation, hosting the Verified Carbon Standard (VCS) programme. VM0042 is Verra's specific methodology for biochar projects. It provides the rules for quantifying the climate benefits of biochar production and application to soil. Published in 2021, VM0042 replaced earlier biochar methodologies and represents the current state of science in biochar carbon accounting.

What VM0042 Covers

VM0042 can be applied to biochar projects that meet the following eligibility criteria:

Criteria	Requirement
Feedstock	Must be waste biomass (crop residue, wood waste, manure). Purpose-grown energy crops are excluded.
Pyrolysis temperature	Must achieve sufficient temperature to produce stable carbon (H:Corg ratio ≤ 0.7)
Biochar application	Must be applied to soil, not used for other purposes (filters, etc.)
Additionality	Biochar production must not be the baseline scenario (common practice test)
Geographic scope	Global — applicable in any country including India

The VM0042 Accounting Framework

VM0042 calculates net GHG reductions from three sources:

- 1. Carbon sequestration in biochar** — the stable carbon locked in the biochar and applied to soil
- 2. Avoided methane and nitrous oxide** — from preventing the open burning or anaerobic decomposition of the feedstock

- 3. Avoided emissions from displaced products** — if biochar replaces fertilisers or lime, the embodied emissions of those products are counted

Minus deductions:

- 4. Project emissions** — energy used in pyrolysis; transport emissions; any N₂O from enhanced soil nitrification
- 5. Leakage** — if the feedstock was previously used for another purpose (animal bedding, etc.), the loss of that use is accounted for

The Permanence Approach: H:Corg Ratio

VM0042 uses the hydrogen-to-organic-carbon (H:Corg) ratio as a proxy for biochar stability. A lower H:Corg ratio indicates greater aromaticity and higher resistance to decomposition. The methodology assigns permanence values based on this ratio:

H:Corg Ratio	Permanence Class	% of Carbon Credited	Typical Production Method
≤ 0.4	Class A (Most Stable)	100% credited	High-temp pyrolysis (>600°C)
0.4–0.6	Class B (Stable)	85% credited	Standard slow pyrolysis (400–600°C)
0.6–0.7	Class C (Moderately Stable)	70% credited	Low-temp pyrolysis (300–400°C)
>0.7	Not eligible	0% credited	Hydrothermal char, char at <300°C

For smallholder TLUD kilns operating at 400–600°C, typically Class B (H:Corg 0.4–0.6). Target 85% credit rate. To qualify for Class A, kiln temperatures should exceed 600°C — achievable with retort or gasifier-type systems.

MRV Requirements for VM0042

Parameter	What to Measure	Frequency	Method
Feedstock volume	Dry weight of biomass processed	Each batch	Calibrated scale
Biochar yield	Dry weight of biochar produced	Each batch	Calibrated scale
H:Corg ratio	Laboratory analysis of biochar samples	Each production batch or annually	Proximate analysis (lab)

Parameter	What to Measure	Frequency	Method
Application rate	Kg biochar applied per hectare	Each application	Field records + GPS coordinates
Pyrolysis temperature	Peak temperature during operation	Each batch	Thermocouple + data logger
Feedstock avoided fate	Documentation that feedstock was waste	Project registration	Supplier declaration

Practical Application: Smallholder Biochar in India

For smallholder biochar projects in India — the most common use case for this methodology — the following project structure is recommended:

- **Aggregation:** Aggregate 200–2,000 smallholders through an FPO or cooperative. VM0042 supports grouped projects.
- **Feedstock:** Rice straw or wheat straw is the most abundant and clearly waste-classified feedstock in North India. Coconut shell in South India.
- **Technology:** TLUD kilns or cone kilns. Both are appropriate. Target H:Corg $\leq$ 0.6.
- **Record-keeping:** Mobile data capture (ODK, Kobo Toolbox) for batch weights, GPS, photographs, and temperature logs.
- **Annual sample laboratory testing:** Send representative biochar samples to a NABL-accredited laboratory annually for H:Corg analysis.
- **Credit value:** At VM0042 Class B (85% credit rate), 1 tonne of biochar produces approximately 2.4–2.8 tCO₂e of credits. At \$25–50/tonne in voluntary markets, each tonne of biochar generates \$60–140 in carbon revenue.